



Illuminating
ENGINEERING SOCIETY

TECHNICAL MEMORANDUM:
PHOTOMETRIC AND ELECTRICAL
MEASUREMENTS OF TUNABLE-WHITE
SOLID-STATE LIGHTING PRODUCTS
AN AMERICAN NATIONAL STANDARD



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Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
IES Testing Procedures Committee**



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CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Normative References	2
2.1	ANSI C78.377-2017	2
2.2	ANSI/IES LS-1-20	2
2.3	CIE 13.3-1995	2
2.4	CIE 15:2018	2
2.5	ANSI/IES TM-27-20	2
2.6	ANSI/IES TM-30-20	2
2.7	ANSI/IES LM-79-19	2
3.0	Definitions	2
3.1	color-control signal	2
3.2	device under test (DUT)	2
3.3	interval format	2
3.4	tunable white	2
4.0	Preparation and Measurement	2
4.1	System	3
4.2	Color-Control Signal	3
4.2.1	Seven or Fewer Levels of Color-Control Signal	3
4.2.2	More Than Seven Levels of Color-Control Signal	3
4.3	Procedure	3
5.0	Data Interpolation	3
6.0	Test Report	4
	Annex A: Interpolation Implications	5
	Annex B: Example Report	6
	References	9

1.0 Introduction and Scope

1.1 Introduction

Lighting products emitting variable spectral power distributions (SPDs) are increasingly prevalent in the architectural lighting marketplace. While a few variable-spectrum products based on fluorescent lamp technology existed in the past, the capabilities of solid-state lighting (SSL) have enabled the development and widespread use of such products. Variable spectral output, however, is not explicitly accounted for in existing methods for characterizing photometric and electrical performance, which are focused on measurements taken at a single operating state, with the product supplied with the intended voltage and without control devices.

The ability to emit radiant power in hundreds or thousands of spectral combinations—only limited by the precision of the control signal being provided—poses a distinct challenge for measuring product performance. Without a default operating state that is indicative of typical performance, and without a practical means to measure performance in every possible operating state, it is necessary to derive a scheme for capturing product performance that balances absolute accuracy with feasibility. This document establishes a common protocol for achieving this goal, relying on the requirements of ANSI/IES LM-79-19 (see **Section 2.7**) for each measurement. It defines the minimum number of measurements to be made and the order in which measurements are to be made. It also provides a framework for data reporting.

To understand the tradeoff between accuracy and feasibility, it is first important to identify what performance characteristics are of interest. The performance of a conventional, fixed-spectrum lighting products is measured when it is operated at the specified supply voltage, and subsequently rated based on luminous flux, power, luminous efficacy, chromaticity, color rendition, power factor, total harmonic distortion, and luminous intensity distribution, among other characteristics. When the spectral power distribution of a product is variable, many of these performance characteristics are also variable; thus, it is necessary to

decide both what characteristics and which level of each characteristic (e.g., minimum, mean, maximum) are important to document. Because true values of these characteristics may not be found with a limited number of measurements, an interpolation procedure can be specified to predict performance at intermediate points.

Products with variable spectral power distributions come in many varieties, each of which may be referred to by several different names. These include products that can emit colored light, products that are restricted to emitting nominally white light, and products that only change spectral power distribution in combination with dimming. Some products may be configurable to operate in more than one way. Each of these varieties presents unique challenges for photometric, colorimetric, and electrical measurement. This Technical Memorandum specifies a procedure of measuring “tunable white” lamps, luminaires, and light engines because it is a subset of variable-spectrum products for which a relatively straightforward measurement protocol can be defined.

1.2 Scope

This document describes a protocol for measuring photometric, colorimetric, and electrical characteristics of tunable-white solid-state lighting products—including lamps, luminaires, and light engines—as covered by ANSI/IES LM-79-19 (see **Section 2.7**). This protocol applies to products for which the spectral power distribution can be adjusted with a single, one-dimensional input having a quantitative, interval format, either continuous or discrete, that is nominally independent of luminous flux control. For example, a product controlled with one variable control input (e.g., slider, rotary knob)—or several presets—for color and one variable control input for lumen output. This document also describes a method for interpolating between measured data to obtain specified characteristics, including correlated color temperature (CCT) range, D_{uv} range, lumen output range (at full intensity control as color changes), efficacy at maximum output, efficacy range, color rendition range (i.e., IES R_i , IES R_g , IES $R_{cs,h1}$, IES $R_{f,h1}$ CIE R_a , CIE R_9), and chromaticity coordinates (i.e., x , y ; u' , v').

The method described in this document does not apply to products that intentionally change chromaticity with luminous flux (e.g., dim-to-warm), or products with multiple color-control input signals (e.g., full-color-tunable) that cannot be set to operate with a single color-control input signal. Some tunable-white products may use control schemes that are not covered by this method, such as a number of categorical presets that are not linked to an underlying scale or a graphical interface that does not provide a quantitative component.

2.0 Normative References

2.1 ANSI C78.377-2017

American National Standard for Electric Lamps—Specifications for the Chromaticity of Solid-State Lighting Products. Rosslyn, Virginia: National Electrical Manufacturers Association. 2017.

2.2 ANSI/IES LS-1-20

Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: Illuminating Engineering Society; 2020. Online: <https://www.ies.org/standards/definitions/>. (Accessed 2021 Feb 11).

2.3 CIE 13.3-1995

Method of Measuring and Specifying Colour Rendering Properties of Light Sources. Vienna: International Commission on Illumination; 1995.

2.4 CIE 15:2018

Colorimetry, 4th ed. Vienna: International Commission on Illumination; 2018.

2.5 ANSI/IES TM-27-20

Technical Memorandum: IES Standard Format for the Electronic Transfer of Spectral Data. New York: IES; 2020.

2.6 ANSI/IES TM-30-20

Technical Memorandum: IES Method for Evaluating Light Source Color Rendition. New York: IES; 2020.

2.7 ANSI/IES LM-79-19

Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products. New York: IES; 2019.

3.0 Definitions

3.1 color-control signal

A signal provided to a driver, via hardware or software, for the purpose of altering a light source's spectral power distribution. This is sometimes referred to as a (*color*) *control channel*. There are several protocols that define communication via these signals, such as Digital Multiplex (DMX), Digital Addressable Lighting Interface (DALI), or 0-10 V. Proprietary communication protocols can also be used.

3.2 device under test (DUT)

Luminaires, lamps, and light engines that operate with solid-state light sources.

3.3 interval format

A type of data format that has a defined order and an equal difference between units. The data may be either discrete or continuous.

3.4 tunable white

Products for which a color-control signal is used to vary the spectral power distribution so that the intended change is in correlated color temperature, measured in kelvins (K). Other names used for this type of product include *white tunable* and *CCT tunable*. As used in this document, *tunable white* does not include products that vary spectrum while nominally holding chromaticity constant (e.g., for tuning melanopic content, or tuning D_{uv} at constant CCT).

4.0 Preparation and Measurement

In all cases, measurements for a given tunable-white product shall be made in an uninterrupted sequence, with the product remaining in a fixed position within a fixed measurement apparatus. The only change

made between measurements shall be the color-control signal. All requirements of ANSI/IES LM-79-19 (refer to **Section 2.7**) shall apply to this protocol, including all aspects of the environmental and electrical test conditions. Accordingly, each measurement will require stabilization; for measurements after the first, the stabilization requirement is only based on rate of change, with no minimum duration (see LM-79).

4.1 System

The tested system shall consist of a device under test (DUT), control hardware, and a user interface with an interval format, such that the input is received on a one-dimensional numerical scale with differences in values being equal. It is important to note that discrete inputs do not need to be equally spaced (e.g., 2700 K, 3000 K, 3500 K), but the scale that underlies those values does (e.g., a 1-K difference in CCT is consistent throughout).

4.2 Color-Control Signal

The process is dependent on the number of discrete control levels that are possible, as described in **Sections 4.2.1** and **4.2.2**.

4.2.1 Seven or Fewer Levels of Color-Control Signal.

For systems with seven or fewer discrete levels at full input power, which is the case for some proprietary controls, all levels shall be measured and reported. To clarify, systems with six or seven discrete levels of color control shall be measured at each level.

4.2.2 More Than Seven Levels of Color-Control Signal. For tunable-white lighting DUTs operated with a color-control input signal allowing for more than seven discrete levels, including nominally continuous control signals, measurements shall be made at five or more levels of the color-control signal provided to the control device via the user interface interval scale. The measurements shall be made at nominally full input power: that is, with any flux-level control set to maximum. The measured input power at each color-control signal may vary.

If possible, the color-control signal input levels should be: minimum, $25\% \pm 5\%$, $50\% \pm 5\%$, $75\% \pm 5\%$, and maximum. (Note that input provided to the control device via the user interface is separate from the

signal from the control device to the light source.) For example, a color control input with a CCT range of 3000 K to 5000 K would be measured at inputs of 3000 K, 3500 K, 4000 K, 4500 K, and 5000 K. Likewise, a system with control input of 0 to 100 would be measured at inputs of 0, 25, 50, 75, and 100. If the specified levels cannot be definitively achieved by the control device via the user interface, five color-control signal levels spaced as equally as possible across the control-signal range should be supplied. When using 8-bit DMX, for example, the target levels would be 0, 64, 128, 191 and 255. In some cases—as may occur with a variable control input (e.g., slider or rotary knob) or graphical interface—the physical input may be non-numerical, but the five levels should still be spaced as equally as possible, with the input documented as described in **Section 6.0**. In all cases, the supplied signals shall be recorded in a manner that supports repeatability.

If the tunable-white lighting DUT can operate without a color-control device, and if the default performance state in absence of a color-control device does not correspond to one of the otherwise measured points, it shall be measured and reported independently.

If interpolated data are not reported (see **Sections 5** and **6**), the DUT shall be measured while operated in the state of maximum power as determined via the specified interpolation procedure.

4.3 Procedure

All measurement for a DUT shall be made according to ANSI/IES LM-79-19. If measurement of the default, control-free condition is needed (as discussed in **Section 4.2.1**), it shall be performed first. Otherwise, it is recommended that the minimum color-control signal be provided first. Subsequent measurements shall be made in ascending order of the color-control signal.

5.0 Data Interpolation

For DUTs operated with a color-control device allowing for more than seven discrete levels, the five measurements for each performance characteristic specified

in **Section 6** may be interpolated using a cubic spline method^{1,2} to a greater number of levels corresponding to the possible color-control signal inputs. For example, measurements for a DUT receiving an 8-bit DMX-based color-control signal could be interpolated to 256 levels, as illustrated in **Annex A**. From these interpolated points, specific important values can be determined and reported, as described in **Section 6**. All interpolated data that is reported shall be noted as such.

It is important to note that, while the data derived from a measured spectral power distribution are interpolated, the measured SPD itself shall not be interpolated.

For DUTs operated with a color-control device allowing for seven or fewer discrete levels, no interpolation shall be performed (because all possible performance states will be measured).

The implications of interpolating the measured data are described in **Informative Annex B**.

6.0 Test Report

The equipment used to set the color-control signals (e.g., app, hardware device) and the color-control signals for measured data shall be reported, as well as the number of possible color-control signal levels, the number of measurements made, and the order of measurement. If numerical data corresponding to the color-control input signal are not available (e.g., with a graphical slider), visual representations shall be provided.

The spectral power distribution (SPD) for each measurement shall be reported in .spd format as specified in IES TM-27-20, and optionally in tabular format, with a wavelength increment of no more than 5 nm (1 nm recommended) and a range no less than 400 to 700 nm (380 to 780 nm recommended).

For each measurement, relevant photometric and electrical characteristics shall be reported, including:

- Luminous flux
- Power consumed

- Luminous efficacy
- CCT
- D_{uv}
- Color rendition, including IES R_f , IES R_g , IES $R_{cs,h1}$, IES $R_{f,h1}$, CIE R_a , and CIE R_9
- Chromaticity coordinates (x, y) and (u', v')

The following shall be reported from the measured dataset and, optionally, from the interpolated data:

- Luminous flux (lumen output) minimum and maximum
- Power consumed minimum and maximum
- Luminous efficacy minimum and maximum
- CCT minimum and maximum
- D_{uv} minimum and maximum
- Color rendition minimum and maximum, including IES R_f , IES R_g , IES $R_{cs,h1}$, IES $R_{f,h1}$, CIE R_a , and CIE R_9

Other performance characteristics (such as power factor) may be included as desired.

In addition to these characteristics, the maximum power within the interpolated data should be determined and performance at that color-control signal reported for: CCT, D_{uv} , lumen output, power, luminous efficacy, IES R_f , IES R_g , IES $R_{cs,h1}$, IES $R_{f,h1}$, CIE R_a , CIE R_9 , and chromaticity coordinates (x, y) and (u', v'). If interpolated data are not reported, the DUT shall be measured while operated in the state of maximum power; this state may be determined via the specified interpolation procedure.

Other characteristics may be reported as desired, following the same interpolation procedure. An example report is provided in **Annex B**. Graphical representation of the measured and interpolated data is recommended. In particular, it is recommended to include charts with the measured data as marked points and interpolated data as lines.

For DUTs with seven or fewer discrete levels, all measured data shall be reported.

Annex A: Interpolation Implications

When measuring the photometric and electrical performance of tunable-spectrum DUTs, there is a tradeoff between expediency of testing and accuracy of the reported results. As shown among several examples in **Figure A-1**, when measuring at five color-control signals, the reported minimum or maximum for a given performance characteristic may be higher or lower, respectively, than the value determined from measurements taken at a greater number of color-control signals. Using a spline interpolation may help to correct this situation, or it could lead to an “overshoot” of the maximum or “undershoot” of the minimum determined from a finer resolution

of actual measurements. (Finding the true minimum or maximum for performance characteristics would require either extensive measurements or highly targeted measurements.)

An analysis of nine products was conducted to examine the tradeoffs associated with interpolation for the 11 performance characteristics identified in this method. The reportable performance characteristics based on five measurements only versus five measurements plus interpolation were compared to baselines determined from measurements taken at 11 color-control signals. The five measurements were subsets of the same 11 measurements for the same DUT, so as not to introduce measurement error into this analysis, which focuses exclusively on procedural differences. All three sets

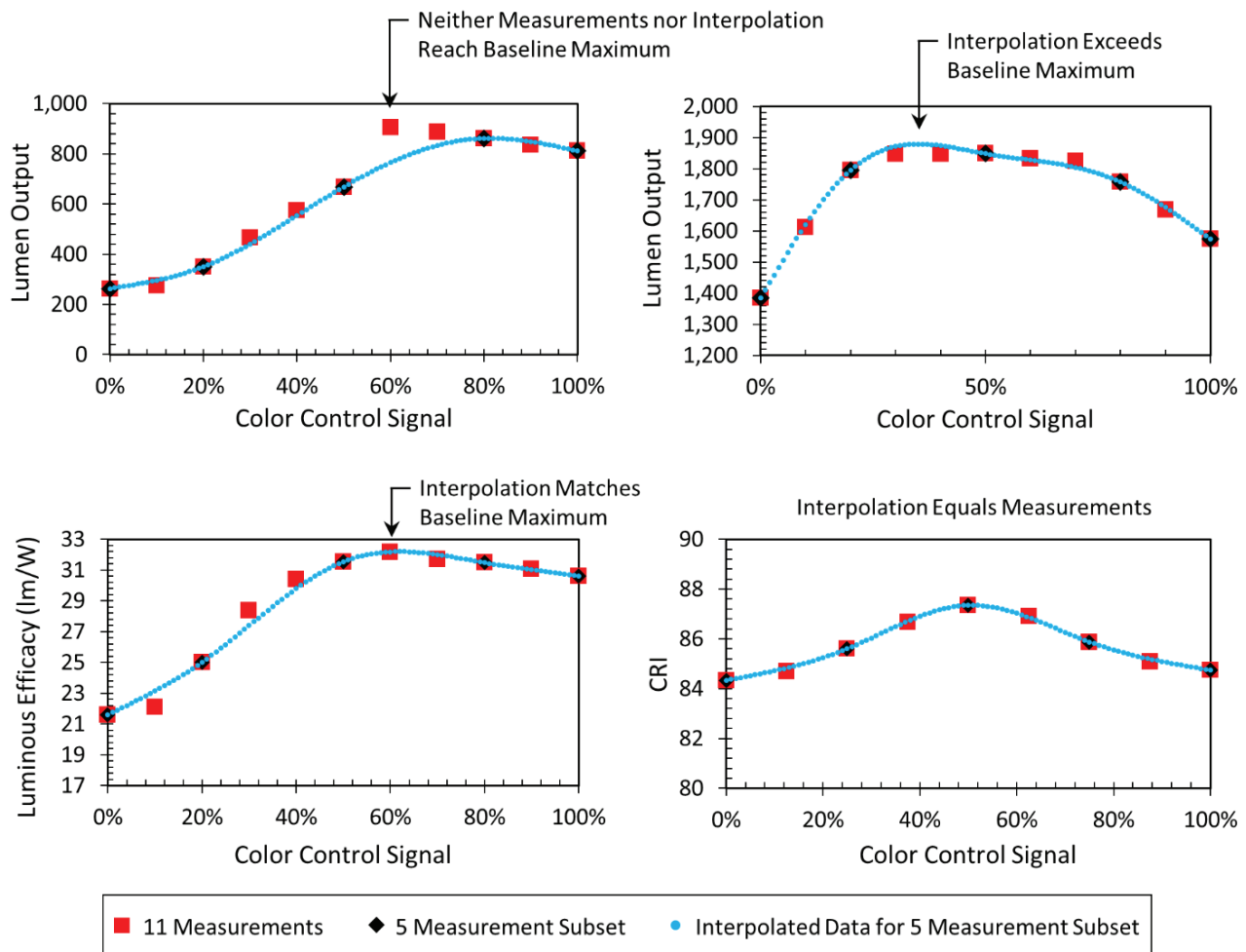


Figure A-1. Possible scenarios when interpolating measured data. The four charts are for different DUTs (and different characteristics). Interpolation may or may not reduce error in minimum and maximum values compared to measured data for a greater number of color-control signal levels.

of data would comply with the methodology in this document. The results can be summarized thus:

- For 4 of 99 maximum-value data points, the maximum interpolated value was greater than the maximum of the baseline measurements by more than 0.1% (0.001 for D_{uv}). This included one value for lumen output (1.7% greater), one value for input power (0.4%), and two values for CIE R_a (0.2%, 0.9%). In all four cases, the maximum values for the five measurements alone were closer to the baseline maximum values, with the lumen output and input power values being the same.
- For 5 of 99 minimum-value data points, the interpolated minimum was less than the measured minimum by more than 0.1% (0.001 for D_{uv}). This included two values for lumen output (-0.1%, -1.7%), one value for luminous efficacy (-1.3%), one value for CCT (-1%), and one value for CIE R_a (-0.6%). In all five cases, the minimum values for the five measurements alone were the same as the baseline minimum values.
- For 20 of 99 maximum-value data points, the maximum values based on five measurements alone were less than the maximum values based on interpolation *and* less than the baseline maximum values by more than 0.1% (0.001 for D_{uv}). The differences between the measured only and measured plus interpolation ranged from -0.2% to -11.3%, whereas the differences between the measured only and baseline maximum values ranged from -0.1% to -21.1%. All values greater in magnitude than -4.5% were for CIE R_g .
- For 2 of 99 minimum-value data points, the minimum values based on five measurements alone were greater than the minimum values based on interpolation *and* greater than the baseline minimum value by more than 0.1% (0.001 for D_{uv}). Both of these cases were for input power (1.0%, 0.6%). In one of these two cases, both the values (measured only and measured plus interpolated) were less than the baseline minimum value.

This anecdotal analysis, as well as **Figure A-1**, shows both the potential improvement in accuracy gained when using interpolation and the downside of sometimes finding values that are outside the baseline range of

performance. In many cases, there is no difference between the alternatives, and in almost all cases the differences are small and likely within manufacturing and/or measurement tolerances.

Five measurement points is the *minimum* required to be in compliance with this method. Errors associated with measuring performance at a small number of color-control signal levels or with interpolation can be reduced by measuring performance at a greater number of color-control signals. If interpolation leads to undesirable results relative to performance thresholds, adding measurements may help to resolve the issue. Omitting interpolation is also acceptable.

Annex B: Example Report

An example of a tunable-white light source test report is provided here. This is strictly informative; specific formatting of included information is at the user's discretion.

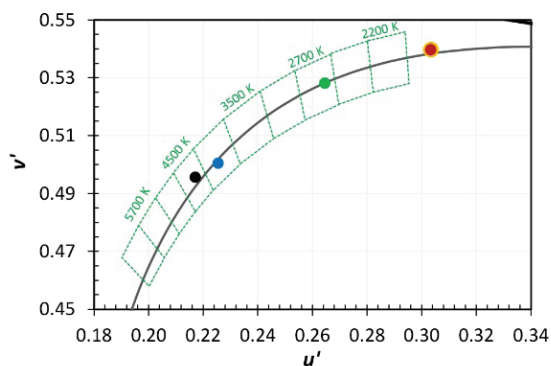
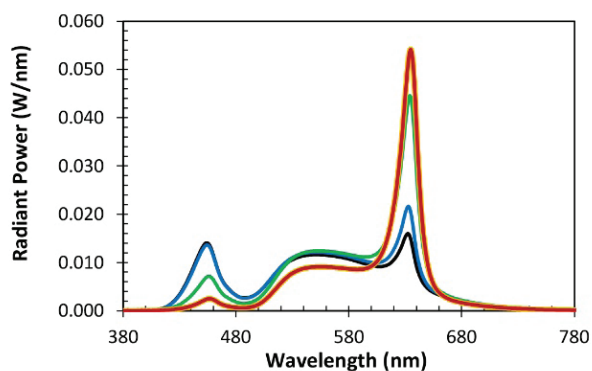
Technical Memorandum: Photometric and Electrical Measurements of Tunable-White Solid-State Lighting Products

IES TM-43-21 Tunable-White Test Report

Test ID: 15-09
 Date: 3/24/2015
 Brand: Example
 Model: Example
 Catalog #: Example
 Control Type: 0-10 V; Numerical interface 0 to 100, increment of 1, 101 total levels
 Make/Model: Example
 Measured Points: 5
 Interpolation? Yes

MEASURED DATA

	Range		Default State	Specified Color Control Signal Targets					Max Power [Interpolated]
	Min	Max		Max	75%	50%	25%	Min	
Meas. Order			1	2	3	4	5	1	
Control Signal			0	100	80	50	20	0	
Output (lm)	703	854	703	720	770	854	703	703	
Power (W)	16.6	19.0	18.4	16.6	17.3	19.0	18.4	18.4	
Efficacy (lm/W)	38.2	45.0	38.2	43.3	44.5	45.0	38.2	38.2	
Power Factor	0.98	0.98	1.0	1.0	1.0	1.0	1.0	1.0	
CCT (K)	2017	4437	2019	4437	4002	2653	2017	2019	
D_{uv}	-0.0008	0.0016	0.0009	0.0016	-0.0008	-0.0001	0.0009	0.0009	
IES R_f	81	87	86	81	83	87	86	86	
IES R_g	99	110	110	99	102	108	110	110	
IES $R_{cs,h1}$	-8%	5%	5%	-8%	-4%	2%	5%	5%	
IES $R_{f,h1}$	85	92	86	85	90	92	86	86	
CIE R_a	83	90	83	84	87	90	83	83	
CIE R_g	51	76	61	51	76	76	61	61	
CIE 1931 x			0.5265	0.3637	0.3797	0.4634	0.5266	0.5265	
CIE 1931 y			0.4163	0.3690	0.3745	0.4111	0.4163	0.4163	
CIE 1976 u'			0.3033	0.2171	0.2255	0.2646	0.3034	0.3033	
CIE 1976 v'			0.5397	0.4956	0.5005	0.5281	0.5397	0.5397	



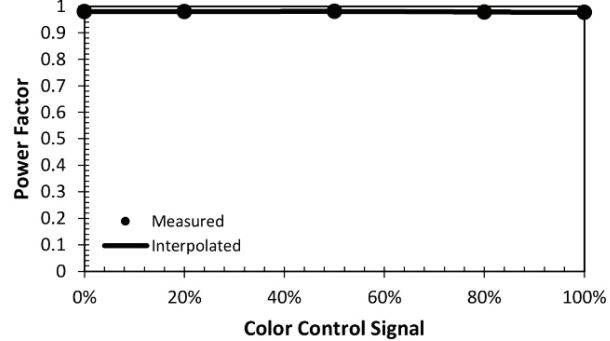
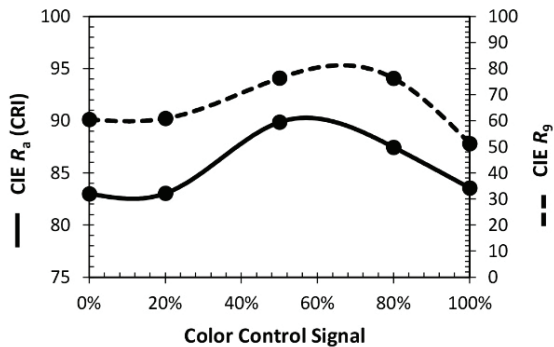
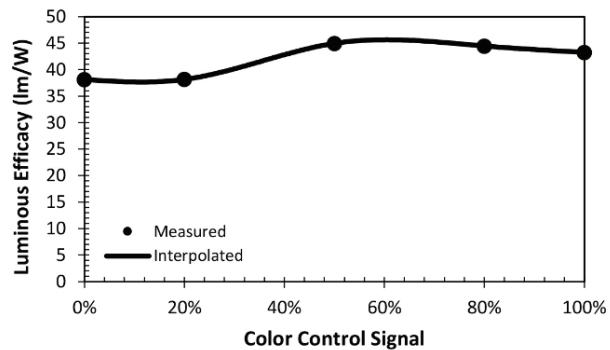
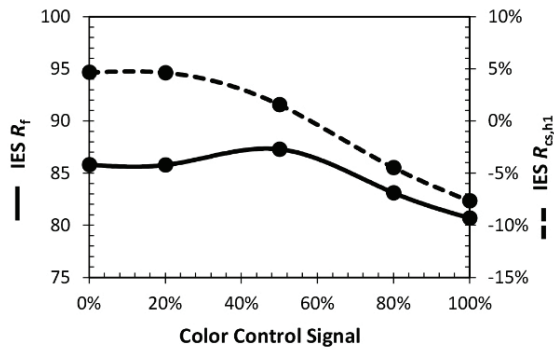
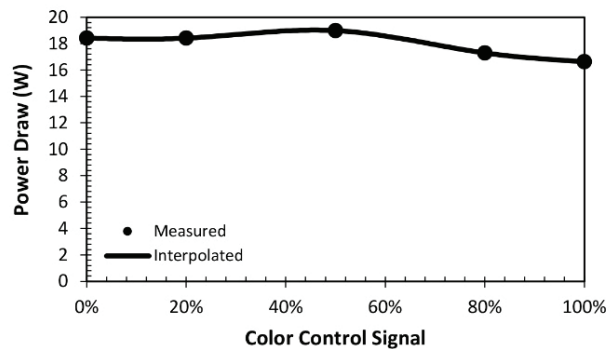
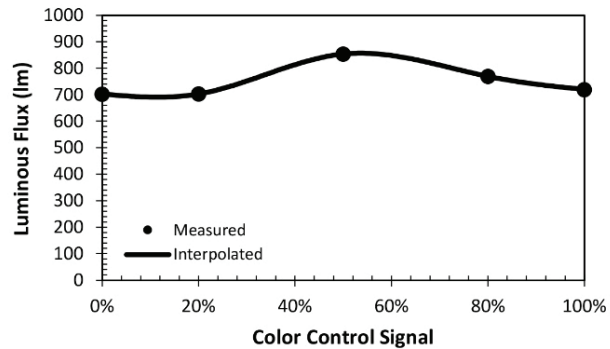
Notes: Negligible SPD change between 0% and 20% control signal.

IES TM-43-21 Tunable-White Test Report

Test ID: 15-09
Date: 3/24/2015

INTERPOLATED DATA

	Range	At Max
	Min Max	Power
Control Signal		46%
Output (lm)	691 – 857	843
Power (W)	16.6 – 19.0	19.0
Efficacy (lm/W)	37.7 – 45.7	44.3
Power Factor	0.98 – 0.98	0.98
CCT (K)	1996 – 4437	2505
D_{uv}	-0.0011 – 0.0016	0.0001
IES R_f	81 – 87	87
IES R_g	99 – 110	109
IES $R_{cs,h1}$	-8% – 5%	2%
IES $R_{f,h1}$	85 – 92	92
CIE R_a	83 – 90	89
CIE R_g	51 – 81	74
CIE 1931 x		0.4753
CIE 1931 y		0.4140
CIE 1976 u'		0.2711
CIE 1976 v'		0.5311



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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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